Introdução Aos Conversores Chaveados

Este trabalho apresenta as principais características dos conversores chaveados, do tipo CC-CC, isto é, conversores que convertem corrente contínua para corrente contínua. Embora este texto esteja na sua primeira versão, os aspectos abordados podem contribuir para a compreensão dos fenômenos envolvidos na conversão CC-CC. Para isso, são apresentados: gráficos, textos, tabelas e equações que tentam descrever os atributos mais significativos dos conversores básicos, cita-se: conversor *Buck* (abaixador), conversor *Boost* (elevador) e conversor *Buck-Boost* (abaixador-elevador).

Como este trabalho encontra-se na sua primeira versão, espera-se que alguns cuidados sejam seguidos para quem for utilizar este material, tais como:

1) pode conter erros nas equações;

2) formas de onda podem estar trocadas ou fora de ordem;

3) a grafia não passou por revisão, logo, erros podem estar contidos nos textos;

Com base nisso, mesmo na presença de erros e/ou incoerências involuntárias, este trabalho pode servir de base para os estudos iniciais na área de Eletrônica de Potência (na qual, engloba vários tipos de conversão de corrente elétrica, tensão elétrica, frequência, fase, amplitude de sinais e potência elétrica) que faz parte dos mais diversos sistemas presentes no dia-a-dia das pessoas.

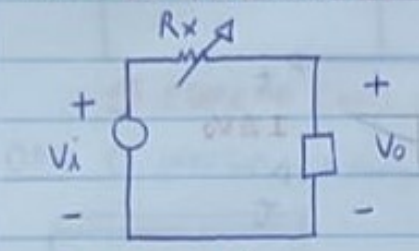
Além disso, na presença de erros, solicita-se que entre em contato por meio de guepfrih@utfpr.edu.br — assim posso providenciar as correções necessárias.

CONVERSORES CHAVEADOS

→ INTRODUÇÃO

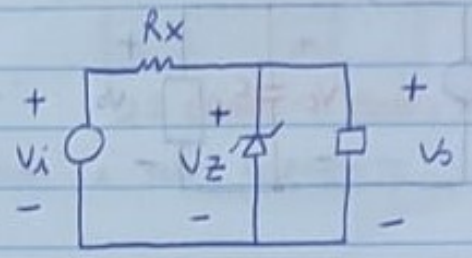
OS CONVERSORES ESTÁTICOS SÃO CARACTERIZADOS POR UTILIZAM, DURANTE O PROCESSAMENTO DE ENERGIA, INTERRUPTORES ELETRÔNICOS QUE JUNTAMENTE COM ELEMENTOS PASSIVOS (INDUTOR, CAPACITOR, RESISTOR, TRANSFORMADOR), A PARTIR DE UMA FONTE PRIMÁRIA, POSSIBILITAM À CARGAS/SISTEMAS OPERAREM DENTRO DE LIMITES ADEQUADOS — SEJA NÍVEIS TENSÃO, FREQUÊNCIA, FASE, POTÊNCIA CORRENTE.

→ TIPOS DE REGULADORES



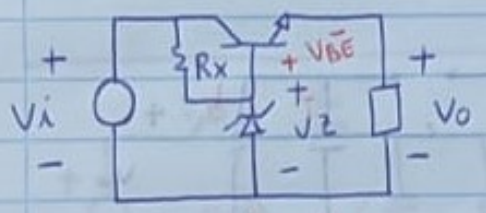
$V_O = V_i \dfrac{R_O}{R_O + R_X}$

$V_{RX} = V_i - V_O$



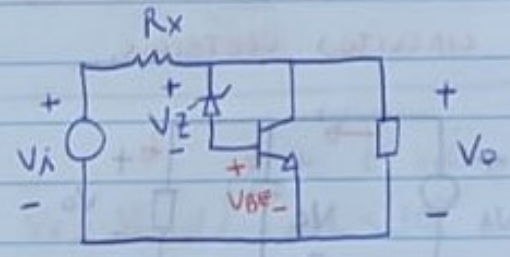
$V_O = V_Z$

$V_{RX} = V_i - V_O$



$V_O = V_Z - V_{BE}$

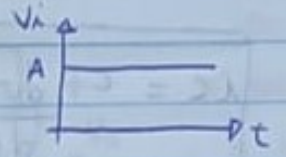
$V_{CE} = V_i - V_O$

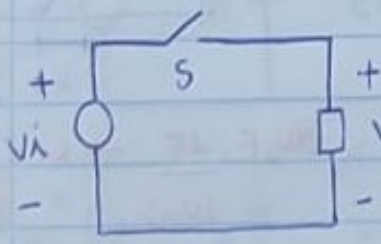


$V_O = V_{CE}$

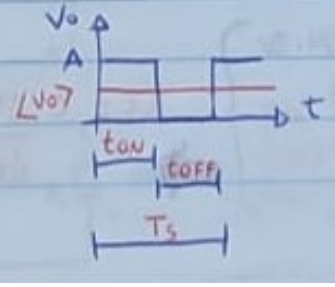
$V_{RX} = V_i - V_O$

$V_O = V_i \cdot d$





S → SWITCH



$T_S = t_{ON} + t_{OFF}$, $d = \dfrac{t_{ON}}{T_S}$ → RAZÃO CÍCLICA

$\langle V_O \rangle$ → Vo MÉDIO

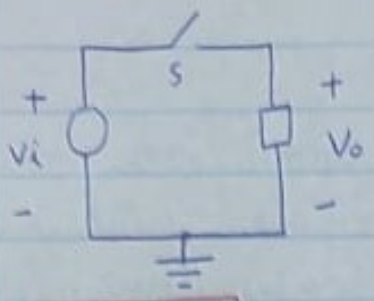
$\langle V_O \rangle = \dfrac{1}{T_S} \int_0^t f(t)\, dt = \dfrac{1}{T_S} \int_0^{t_{ON}} A\, dt$

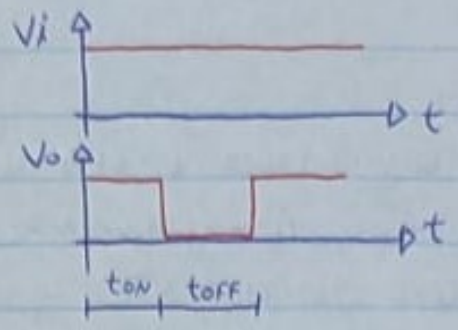
$\langle V_O \rangle = \dfrac{1}{T_S} A \Big|_0^{t_{ON}} \rightarrow \langle V_O \rangle = \dfrac{A}{T_S} (t_{ON} - 0)$

$\langle V_O \rangle = A\, t_{ON}/T_S \rightarrow \langle V_O \rangle = V_i \cdot d$

→ DERIVAÇÃO DO CONVERSOR BUCK (MCC)

(A)





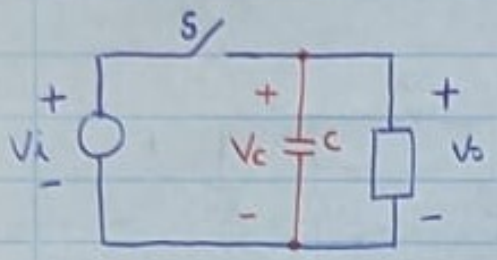
$$\begin{cases} T_s = t_{ON} + t_{OFF} \\ d = \dfrac{t_{ON}}{T_s} \\ f_s = \dfrac{1}{T_s} \end{cases}$$

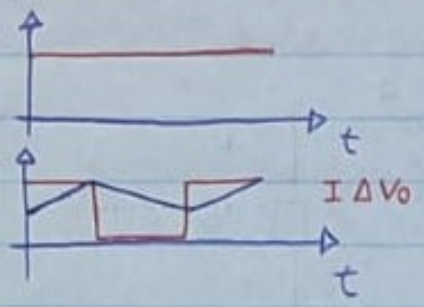
$$\boxed{V_o = V_i \cdot d}$$

$T_s = d + t_{OFF} \rightarrow t_{OFF} = T_s - \dfrac{t_{ON}}{T_s} \rightarrow$ ~~$t_{OFF} = \dfrac{T_s}{T_s} - \dfrac{t_{ON}}{T_s}$~~

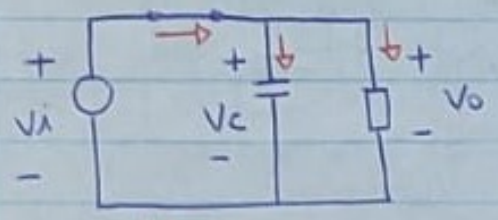
~~$t_{OFF} = (1-d)$~~ $t_{OFF} = T_s - d \cdot T_s \rightarrow t_{OFF} = (1-d)\,T_s$

(B)

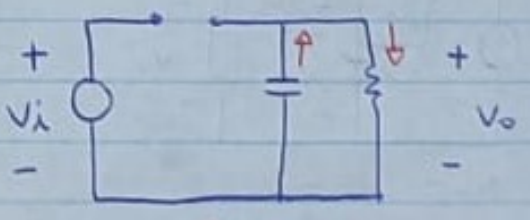




* ESSE CIRCUITO PERMITE MANTER A TENSÃO DE SAÍDA
* PORÉM, COM O INTERRUPTOR EM CONDUÇÃO → COLOCA DUAS FONTES DE TENSÃO EM PARALELO → FERE OS PRINCÍPIOS BÁSICOS DE CIRCUITOS ELÉTRICOS.

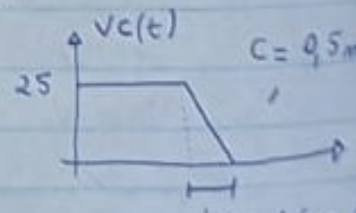


1ª ETAPA



2ª ETAPA

$$\boxed{i_C = C\,\frac{dv_C}{dt}}$$

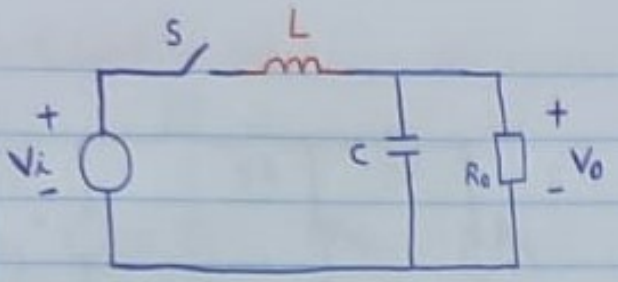


OBS: CAPACITOR NÃO PERMITE VARIAR A TENSÃO BRUSCAMENTE

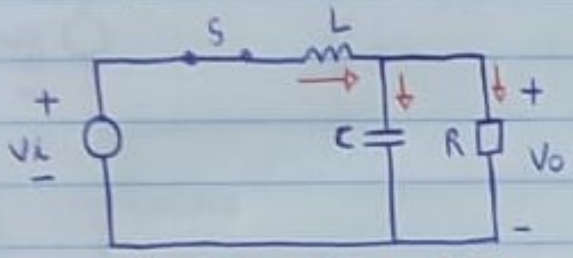
$$i_C = C\,\frac{\Delta V}{\Delta t} = 500\mu F \cdot \frac{25}{10nS} \rightarrow i_C = 1{,}25\ MA$$

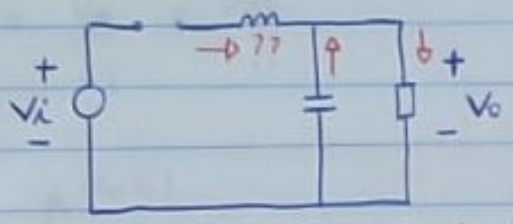
MEGA AMPER

(C)



* ESTE CIRCUITO PERMITE MANTER A TENSÃO DE SAÍDA
* NÃO DEIXA DUAS FONTES DE TENSÃO (V_A e V_C) SEREM CONECTADAS EM PARALELO (PRINCÍPIOS BÁSICOS DE CIRCUITOS ELÉTRICOS
* PORÉM, NÃO OFERECE UM CAMINHO PARA A CORRENTE DO INDUTOR DURANTE O BLOQUEIO DO INTERRUPTOR

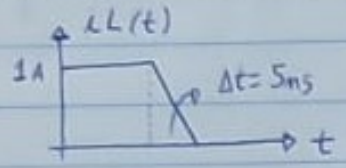




1ª ETAPA → $t_{ON} = d \cdot T_s$ 2ª ETAPA → $t_{OFF} = (1-d)\,T_s$

OBS: O INDUTOR NÃO PERMITE VARIAR A SUA CORRENTE BRUSCAMENTE

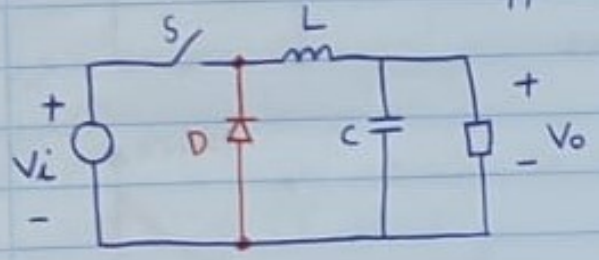
$$v_L = L\frac{di_L}{dt}$$



$L = 2mH$

$$v_L = L\frac{di_L}{dt} \rightarrow v_L = 2m\,\frac{1A}{5ms} \rightarrow v_L = 400\ kV$$

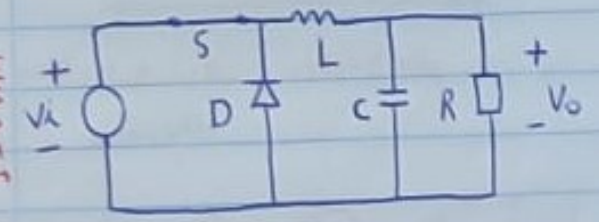
(D)



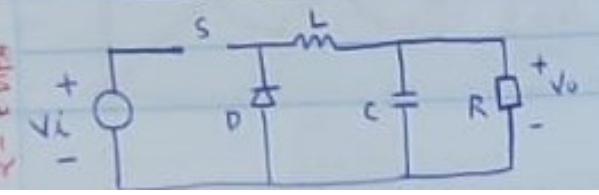
$V_o = V_i \cdot d$ → SÓ É VÁLIDO SE $\Delta i_L < 30\%$ e $\Delta V_c < 30\%$

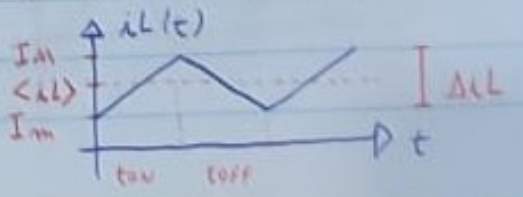
* ESTE CIRCUITO PERMITE UM CAMINHO PARA A CORRENTE DO INDUTOR QUANDO O INTERRUPTOR ENTRA EM BLOQUEIO
* MODO DE CONDUÇÃO CONTÍNUA (MCC) A CORRENTE NO INDUTOR NÃO CHEGA A ZERO DURANTE AS ETAPAS DE OPERAÇÃO

1ª ETAPA

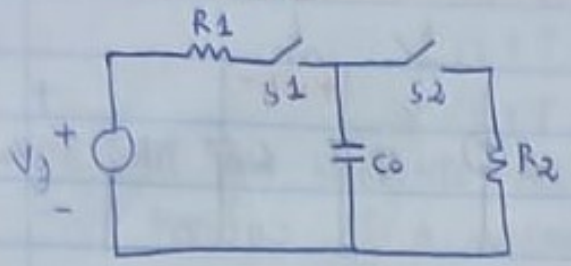


2ª ETAPA





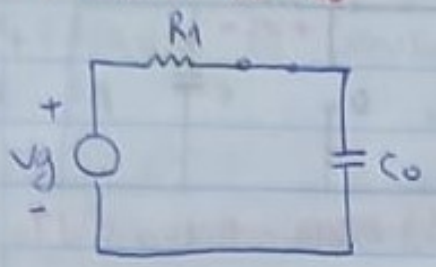
→ CIRCUITO RC



$i = dq/dt \qquad \Delta Q = i \cdot \Delta t$

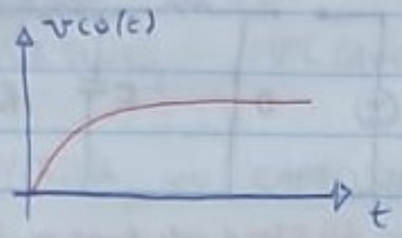
$v_C = q/C \rightarrow \Delta Q = C \cdot \Delta V$

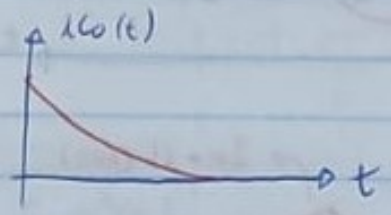
→ CARGA DE Co



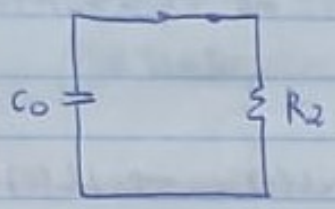
$$v_{Co}(t) = V_g \left(1 - e^{-t/RC}\right)$$

$$i_{Co}(t) = \frac{V_g}{R_1} e^{-t/RC}$$



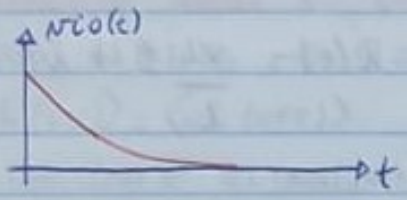


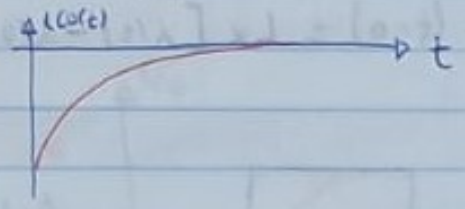
→ DESCARGA DE Co



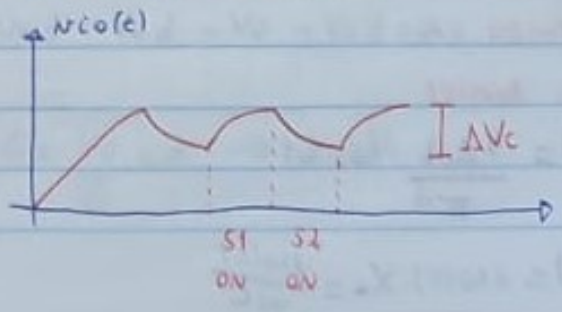
$$v_{Co}(t) = V_g e^{-t/RC}$$

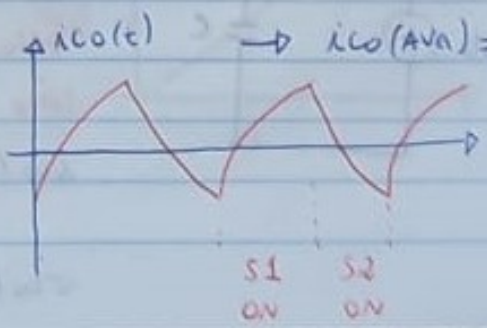
$$i_{Co}(t) = -\frac{V_g}{R_2} e^{-t/RC}$$





→ CIRCUITO CHAVEADO





→ $i_{Co}(AVG) = 0$

→ SINAL PERIÓDICO

$$i(t+T) = i(t)$$
$$v(t+T) = v(t)$$

OBS: P/ UMA CORRENTE PERIÓDICA, A CORRENTE AO FINAL DE UM PERÍODO É A MESMA QUE NO INÍCIO. O MESMO É VÁLIDO PARA UMA TENSÃO PERIÓDICA.

→ ENERGIA NO INDUTOR

$$w(t) = \frac{1}{2} L i^2(t)$$

PARA UMA CORRENTE PERIÓDICA → $W_{LÍQUIDA} = 0$

LOGO, $P_L = 0$

PORTANTO, $v_L(AVG) = 0$

→ ENERGIA NO CAPACITOR

$$w(t) = \frac{1}{2} C v^2(t)$$

PARA UMA TENSÃO PERIÓDICA → $W_{LÍQUIDA} = 0$

LOGO, $P_C = 0$

PORTANTO, $i_C(AVG) = 0$

→ TENSÃO MÉDIA NO INDUTOR → $v_L = 0$

→ CORRENTE MÉDIA NO CAPACITOR → $i_C = 0$

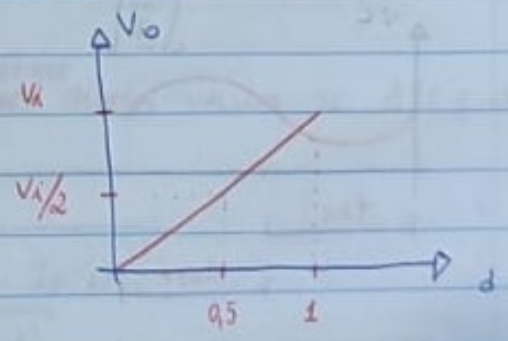
→ GANHO ESTÁTICO → $G = V_o/V_i$

$\langle v_L(ON) \rangle = \langle v_L(OFF) \rangle$

$v_L(AVG)(ON) = v_L(AVG)(OFF)$

$(V_i - V_o)\, d\, T_s = V_o (1-d) T_s$

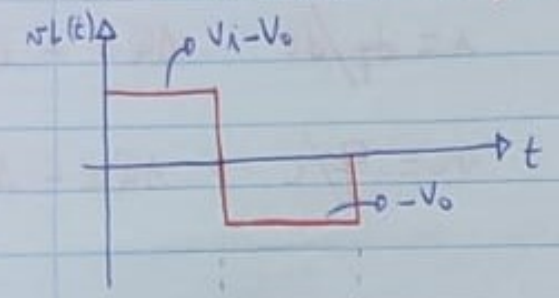
$V_i d - V_o d - V_o + V_o d = 0$

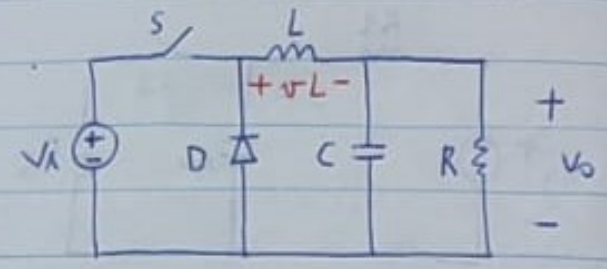


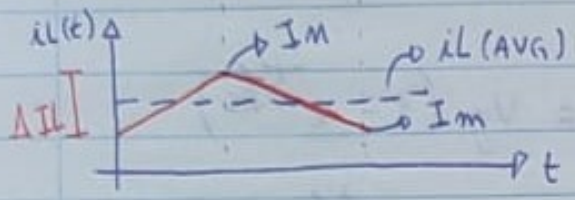
$G = \frac{V_o}{V_i} = d \rightarrow V_o = V_i . d$

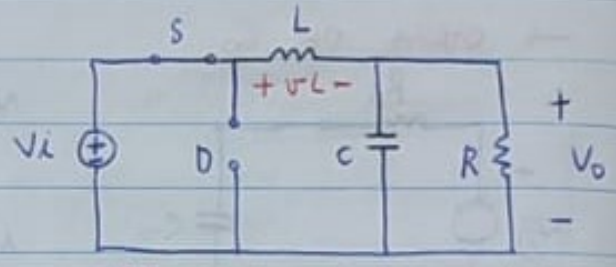
A

→ CONVERSOR BUCK ~~TCM~~ ~~MODOS DE OPERAÇÃO~~ (CCM)

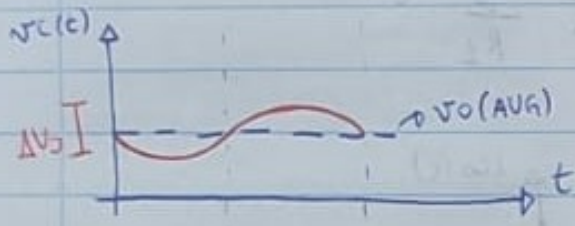


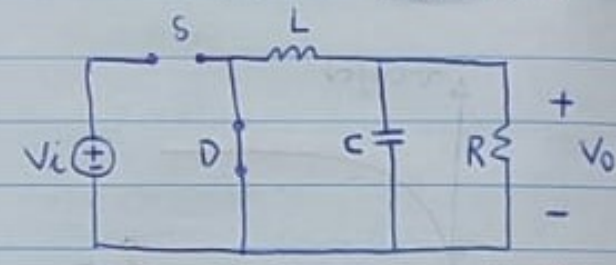




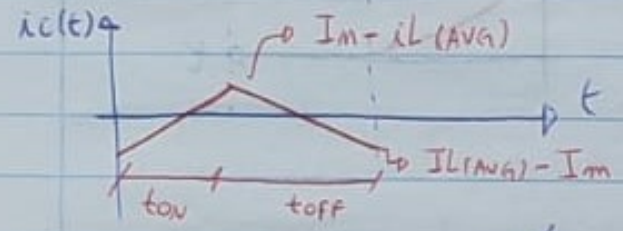


1ª ETAPA → $t_{ON} = d\,T_s$





2ª ETAPA → $t_{OFF} = (1-d)\,T_s$

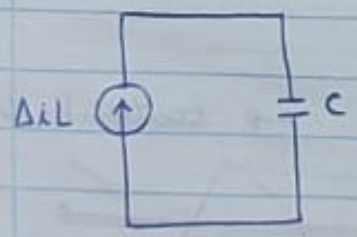


→ $v_L = L\dfrac{di(t)}{dt}$ → $\displaystyle\int_0^t v_L\,dt = L\int_0^t \frac{di(t)}{dt}\,dt$ → ~~$v_L = L\,i(t-0)$~~

~~$v_L(t-0) = L\,i(t)$ → $v_L(t) = L\,i_L(t)$ → $i_L(t) = \dfrac{v_L\,t}{L}$~~

$v_L(t-0) = L \times [i(t) - i(0)]$ → $i(t) = \dfrac{v_L}{L}\,t + i(0)$ (*)

→



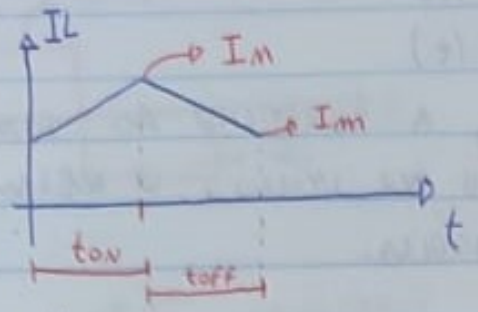
$\Delta i_L = i_{co}$

DECOMPONDO EM SÉRIE DE FOURIER

$i_{co}(t) = \dfrac{4\,\Delta i_L}{\pi^2}\cos \omega t$

$v_{co}(t) = i_{co}(t) \cdot X_c = \dfrac{i_{co}(t)}{\omega C}$

→ ESPECIFICAÇÃO DOS ELEMENTOS PASSIVOS

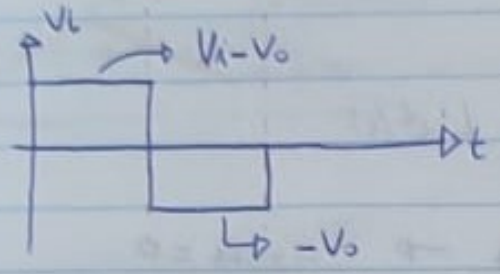
→ 

$$v_L = L\frac{di_L}{dt} \qquad T_s = \frac{1}{f_s}$$

$$v_L = L\frac{\Delta i_L}{\Delta t}$$

$$t_{ON} \rightarrow d\,T_s$$

$$t_{OFF} \rightarrow (1-d)\,T_s$$

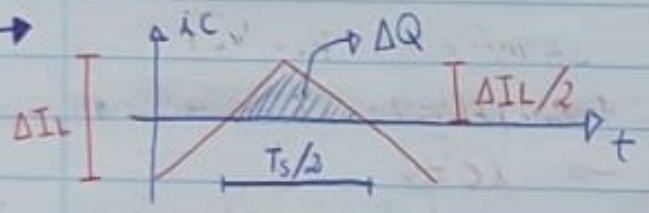


$$\frac{V_i - V_o}{L} = \frac{\Delta i_L}{d\,T_s} \rightarrow \Delta i_L = \frac{(V_i - V_o)\,d\,T_s}{L\,f_s}$$

$$i(t) = \frac{v_L}{L}\,t + i(0) \qquad L = \frac{(V_i - V_o)\,d}{f_s\,\Delta I}$$

$i(0) = I_m$; $t = t_{ON}$; $i(t) = I_M$; $v_L = (V_i - V_o)$

$$I_M = \underbrace{\frac{(V_i - V_o)}{L}\,d\,T_s}_{\Delta I_L} + I_m \rightarrow \Delta I_L = I_M - I_m$$

→ 

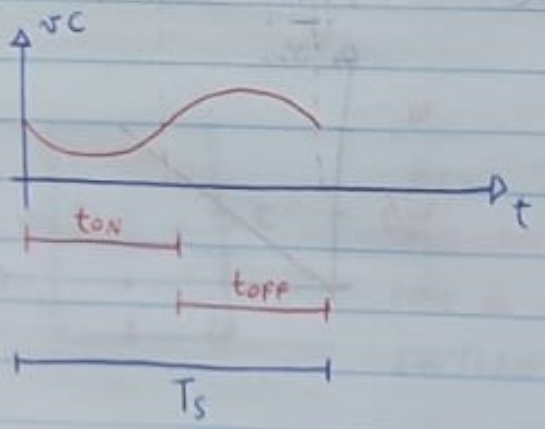
$\cancel{\rightarrow = f_i \cdot \Delta t}$ $\quad i = dq/dt$

$$\Delta Q = C\,\Delta V$$

DEFINIR A AREA HACHURADA

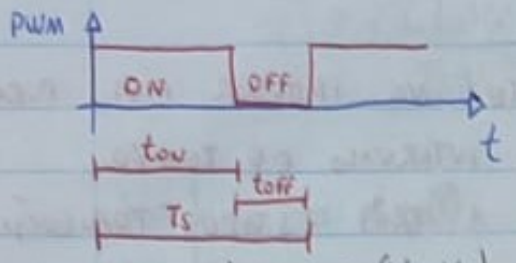
ÁREA DO TRIÂNGULO → $\Delta Q = \frac{B \cdot h}{2}$

$$\Delta Q \Rightarrow \frac{1}{2}\,\frac{T_s}{2}\,\frac{\Delta I}{2} = C\,\Delta V$$



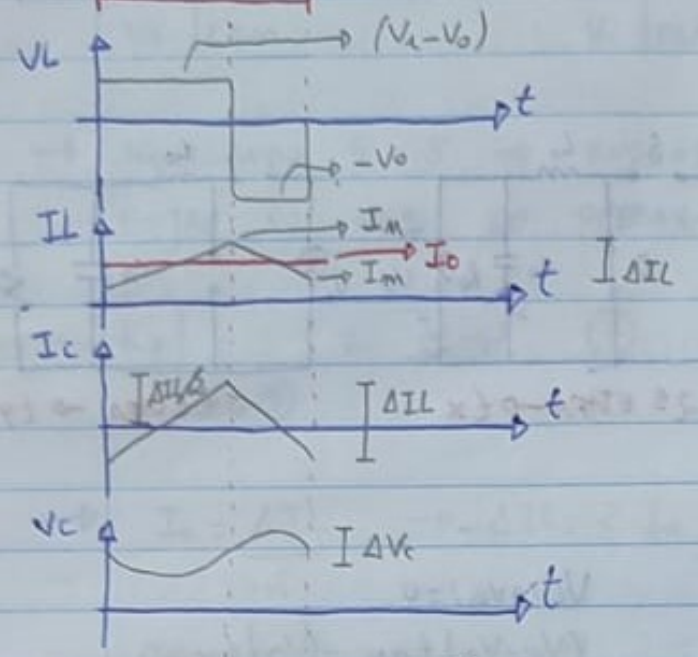
$$C = \frac{\Delta I}{8\,f_s\,\Delta V} = \frac{V_o\,(1-d)}{8\,L\,f_s^2\,\Delta V}$$

→ PRINCIPAIS FORMAS DE ONDA

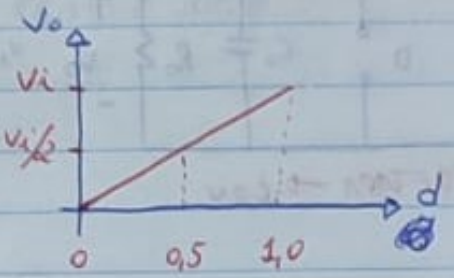


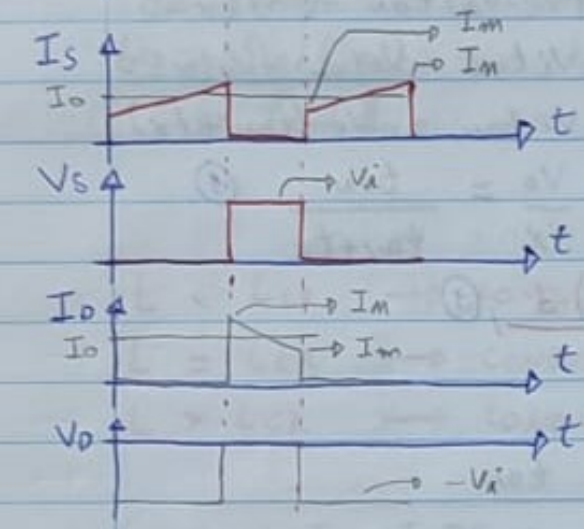
$$T_S = \frac{1}{f_S} = t_{ON} + t_{OFF}$$

$$t_{ON} = d.T_S \quad , \quad t_{OFF} = (1-d).T_S$$



$$V_o = d.V_i$$
$$0 \leq d \leq 1$$





$$\begin{cases} \Delta I_L = I_M - I_m \\ I_M = I_o + \Delta I_L/2 \\ I_m = I_o - \Delta I_L/2 \end{cases}$$

$$\Delta I_L = \frac{(V_i - V_o)\,d}{L_o\, f_S}$$

$I_o = V_o/R_o$; $P_o = V_o I_o$

$$I_{S(AVG)} = \frac{1}{T_S}\int_0^{t_{ON}} I_o\,dt \rightarrow I_{S(AVG)} = d.I_o$$

$$I_{S(RMS)} = \sqrt{\frac{1}{T_S}\int_0^{t_{ON}} I_o^2\,dt} \rightarrow I_{S(RMS)} = \sqrt{d}.I_o$$

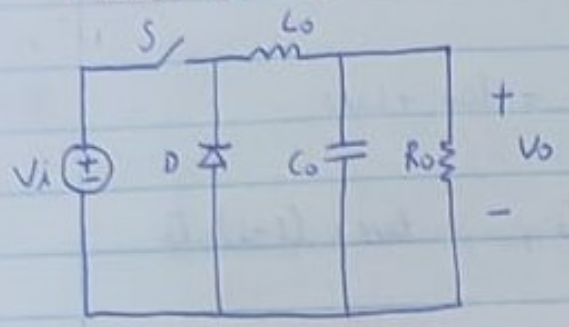
$$I_{D(AVG)} = (1-d).I_o$$
$$I_{D(RMS)} = \sqrt{1-d}.I_o$$

OBS: EQUAÇÕES DOS VALORES EFICAZES VÁLIDAS SE $\Delta I_L \leq 20\%$

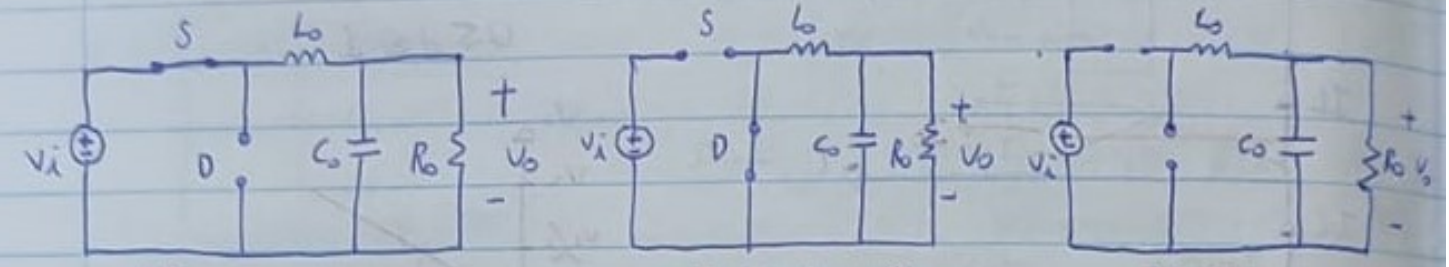
$$V_{S(MAX)} = V_i$$
$$V_{D(MAX)} = |V_i|$$

$$I_{CO} = \sqrt{I_{D(RMS)}^2 - I_{D(AVG)}^2} = I_o\sqrt{d(1-d)}$$

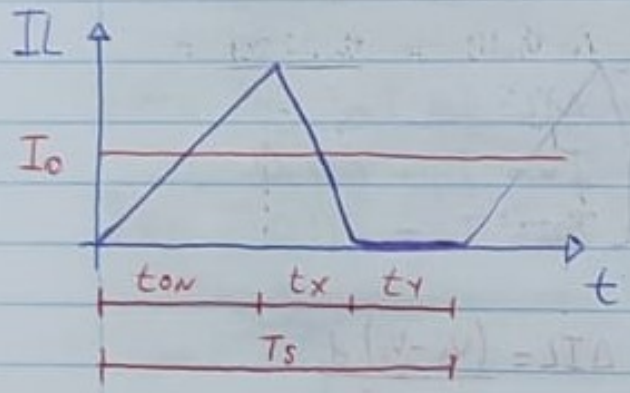
→ MODO DE CONDUÇÃO DESCONTÍNUA (DCM)



* A CORRENTE NO INDUTOR FICA DURANTE UM CERTO INTERVALO DE TEMPO.
* ISSO LEVA A TRÊS ESTADOS TOPOLÓGICOS



1ª ETAPA → t_{ON} 2ª ETAPA → t_X 3ª ETAPA → t_Y



$V_{L(AVG)} = 0$

$(V_i - V_o)\, t_{ON} - (V_o)\, t_X = 0$

$V_i t_{ON} - V_o t_{ON} - V_o t_X = 0$

$V_i t_{ON} = V_o (t_{ON} + t_X)$

$$\frac{V_o}{V_i} = \frac{t_{ON}}{t_{ON} + t_X} \quad (1)$$

$$I_o = \frac{1}{T_s} \frac{I_M (t_{ON} + t_X)}{2}, \quad (2) \qquad I_M = \frac{(V_i - V_o)\, d}{L_o f_s}, \quad (3)$$

$$t_X = \frac{2 I_o T_s - I_M t_{ON}}{I_M}; \qquad \frac{V_o}{V_i} = \frac{t_{ON}}{t_{ON} + \frac{2 I_o T_s - I_M t_{ON}}{I_M}}$$

$$V_o = \frac{V_i^2 d \cdot t_{ON}}{2 I_o L_o + V_i d\, t_{ON}} \rightarrow V_o = \frac{T_s V_i^2 d^2}{T_s V_i d^2 + 2 I_o L_o} \doteq T_s V_i^2$$

$$V_o = \frac{V_i d^2}{d^2 + \frac{2 I_o L_o V_i}{T_s V_i^2}} \rightarrow \frac{V_o}{V_i} = \frac{d^2}{d^2 + K} \; ; \; K = \frac{2 I_o L_o}{T_s V_i}$$

$$\left.\frac{V_o}{V_i}\right|_{CCM} = d \; ; \; (4) \qquad \left.\frac{V_o}{V_i}\right|_{DCM} = \frac{d^2}{d^2 + K} \; ; \; (5) \quad K = \frac{2 I_o L_o}{T_s V_i} \; (6)$$

→ IGUALANDO 4 e 5 → ENCONTRA-SE A REGIÃO DE FRONTEIRA (K_F) ENTRE OS MODOS DE OPERAÇÃO CCM E DCM.

$$\left.K_F\right|_{(4)=(5)} = d - d^2 \quad (7)$$

$$\rightarrow \; I_o = \frac{\Delta IL}{2} \rightarrow \Delta IL = 2 I_o = \frac{(V_i - V_o) d}{f_s L_o} = \frac{V_o (1-d)}{f_s L_o} =$$

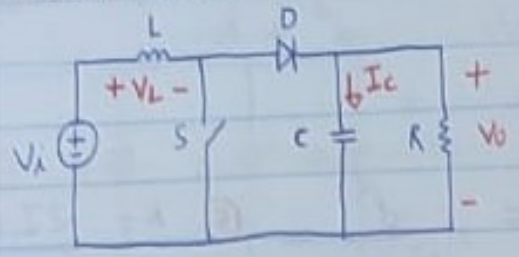
$$L_{CR} = \frac{(V_o)(1-d)}{f_s 2 I_o} \rightarrow L_{CR} = \frac{R_o (1-d)}{2 f_s}$$

$L < L_{CR}$ → CONDUÇÃO DESCONTÍNUA

$L = L_{CR}$ → CONDUÇÃO CRÍTICA

$L > L_{CR}$ → CONDUÇÃO CONTÍNUA

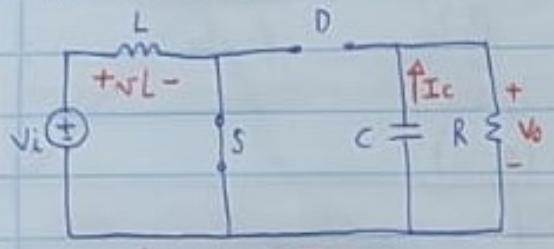
→ CONVERSOR BOOST



* CONVERSOR ELEVADOR
* TENSÃO DE SAÍDA (V_o) É MAIOR QUE A TENSÃO DE ENTRADA (V_i) ($V_o \geq V_i$)
* A ANÁLISE CONSIDERA APENAS O MODO DE CONDUÇÃO CONTÍNUA (CCM)

* EM CCM APRESENTA DOIS ESTADOS TOPOLÓGICOS, QUE ESTÃO ASSOCIADOS À CONDUÇÃO E O BLOQUEIO DOS SEMICONDUTORES — DIODO E INTERRUPTOR — QUE OCORRE DE MANEIRA COMPLEMENTAR. ISSO LEVA A DOIS CIRCUITOS EQUIVALENTES E ~~SUAS~~ AS RESPECTIVAS FORMAS DE ONDA DOS COMPONENTES.

→ PRIMEIRA ETAPA DE OPERAÇÃO (INTERRUPTOR EM CONDUÇÃO)



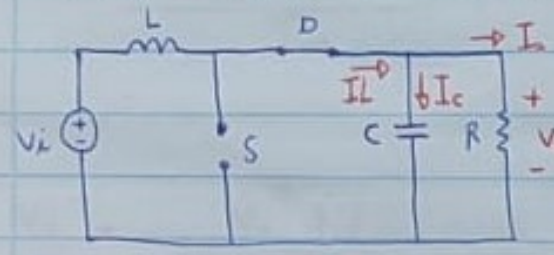
$t_{ON} \rightarrow d\,T_S$

* INTERRUPTOR EM CONDUÇÃO → S (ON)
* DIODO REVERSAMENTE POLARIZADO → D (OFF)
* INDUTOR SENDO MAGNETIZADO → ARMAZENA ENERGIA
* CAPACITOR SENDO DESCARREGADO → ENTREGA ENERGIA

$$v_L = V_i \rightarrow v_L = L\frac{di_L(t)}{dt} \rightarrow \frac{di_L(t)}{dt} = \frac{V_i}{L}$$

$$i_C = -I_o \rightarrow i_C = C\frac{dv_C(t)}{dt} \rightarrow \frac{dv_C(t)}{dt} = -\frac{I_o}{C}$$

→ SEGUNDA ETAPA DE OPERAÇÃO (INTERRUPTOR EM BLOQUEIO)



* INTERRUPTOR EM BLOQUEIO → S (OFF)
* DIODO DIRETAMENTE POLARIZADO → D (ON)
* INDUTOR É DESMAGNETIZADO → ENTREGA ENERGIA
* CAPACITOR É CARREGADO → ARMAZENA ENERGIA

$$v_L = V_i - V_o \rightarrow v_L = L\frac{di_L(t)}{dt} \rightarrow \frac{di_L(t)}{dt} = \frac{V_i - V_o}{L}$$

$$i_C = I_L - I_o \rightarrow i_C = C\frac{dv_C(t)}{dt} \rightarrow \frac{dv_C(t)}{dt} = \frac{I_L - I_o}{C}$$

→ GANHO ESTÁTICO (G)

$$G \triangleq \frac{V_o}{V_i}$$

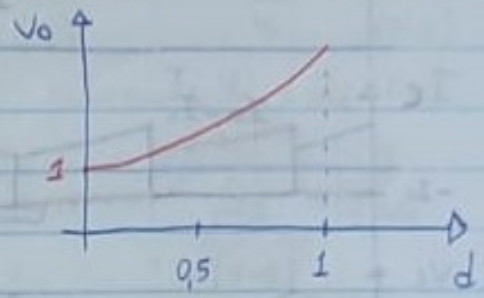
CONSIDERANDO O PRINCÍPIO DO BALANÇO VOLT-SEGUNDO NO INDUTOR CHEGA-SE NA SEGUINTE CONDIÇÃO → $v_L = 0$ EM UM PERÍODO DE COMUTAÇÃO. LOGO,

$$v_L(ON) + v_L(OFF) = 0$$

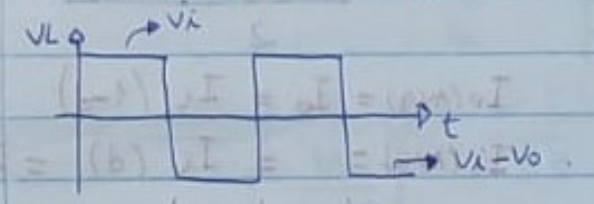
$$V_i(d)T_s + (V_i - V_o)(1-d)T_s = 0$$

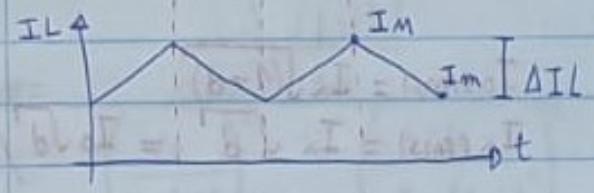
$$\cancel{V_i(d)} + V_i - \cancel{V_i(d)} - V_o(1-d) = 0$$

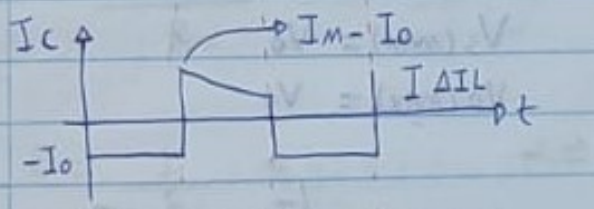
$$\boxed{\frac{V_o}{V_i} = \frac{1}{1-d}}$$

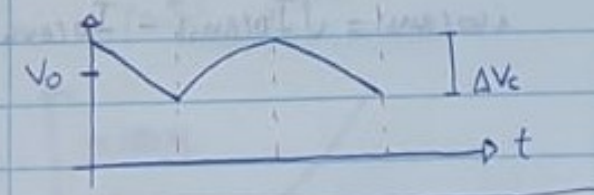


→ PROJETO DOS COMPONENTES PASSIVOS









* DA PRIMEIRA ETAPA DE OPERAÇÃO

$$v_L = L\frac{di_L}{dt} \rightarrow V_i = L\frac{\Delta i_L}{\Delta t}$$

$$\boxed{L = \frac{V_i\, d}{f_s\, \Delta I_L}}$$

$$i_C = C\frac{dv_C}{dt} \rightarrow I_o = C\frac{\Delta V_c}{\Delta t}$$

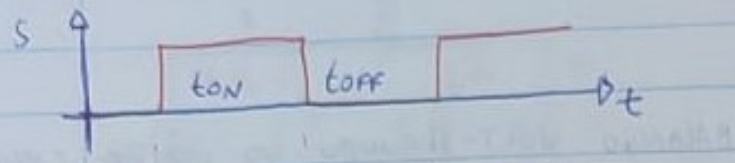
$$\boxed{C = \frac{I_o\, d}{f_s\, \Delta V_c}}$$

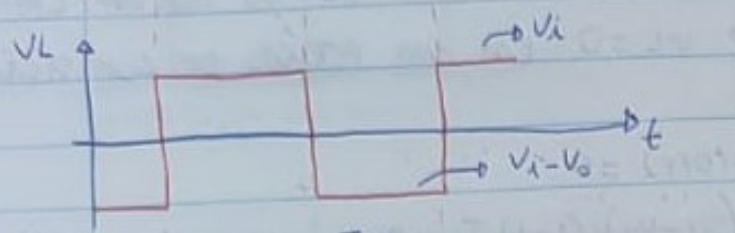
$$0 = \frac{1}{T_s}\left[v_L(ON)\, dT_s + v_L(OFF)(1-d)T_s\right]$$

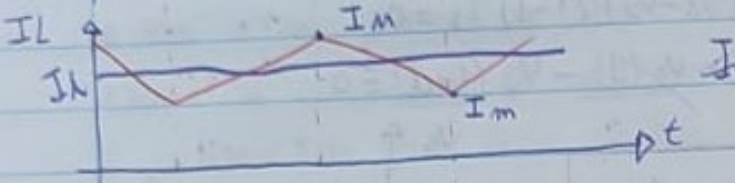
$$v_L = 0 \rightarrow \frac{1}{T_s}\left[\int_0^{t_{ON}} v_L(ON)\,dt + \int_{t_{ON}}^{T_s} v_L(OFF)\,dt\right]$$

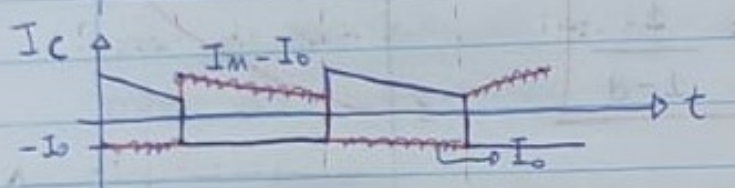
$$v_L(OFF) = v_L(ON)\frac{d}{d-1}$$

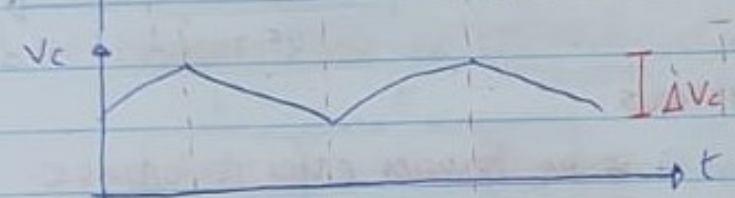
→ PRINCIPAIS FORMAS DE ONDA

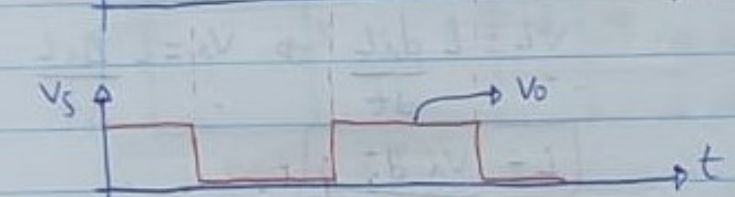


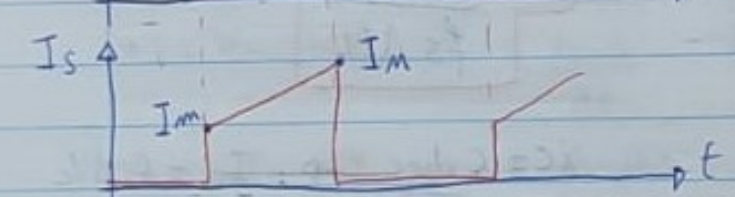


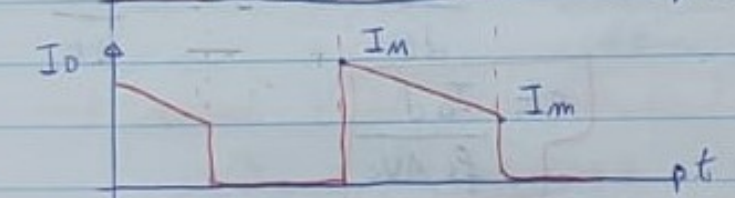


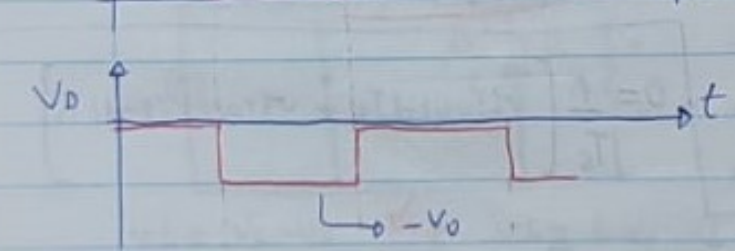












$P_i = P_o$

$V_i . I_i = V_o . I_o$

$I_i = \frac{V_o}{V_i} . I_o \rightarrow I_i = \frac{I_o}{1-d}$

$I_{L(AVG)} = I_i = \frac{I_o}{1-d} = \frac{V_i}{R_o (1-d)^2}$

$\cancel{I_{D(AVG)} I_o = \frac{1}{T_S} \int_{t_{ON}}^{T_S} \left[I_m - \frac{(V_o - V_i)}{L} t \right] dt}$

$I_o = I_{D(AVG)} = \frac{1}{T_S} \int_0^{t_{OFF}} \frac{I_m - I_m}{t_{OFF}} t + I_m dt$

$I_{D(AVG)} = \frac{(I_M + I_m)(1-d)}{2}$

$I_{O(AVG)} = I_o = I_i (1-d)$

$I_{S(AVG)} = \quad = I_i (d) = I_o d / (1-d)$

$I_{D(RMS)} = I_i \sqrt{(1-d)}$

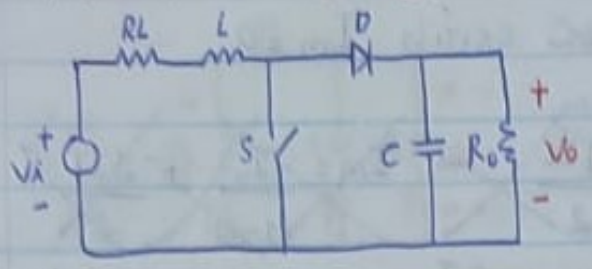
$I_{S(RMS)} = I_i \sqrt{d} = I_o \sqrt{d} / (1-d)$

$V_{S(MAX)} = V_o$

$V_{D(MAX)} = V_o$

$i_{CO(RMS)} = \sqrt{I_{D(RMS)}^2 - I_{O(AVG)}^2}$

→ RENDIMENTO DO CONVERSOR



$$\eta = \frac{P_o}{P_o + P_{PERDAS}}$$

$P_{LOSS} = I_i^2 \cdot RL$; $\frac{V_o}{V_i} = \frac{1}{(1-d)}$; SE → $P_i = P_o$

$V_i \cdot I_i = V_o \cdot I_o$

$I_i = I_o \frac{V_o}{V_i} = \frac{I_o}{(1-d)}$

$I_i = I_o \frac{1}{(1-d)} = \frac{V_o}{R_o} \frac{1}{(1-d)} = \frac{V_i}{R_o (1-d)^2}$

$I_o = I_i \frac{V_i}{V_o} = I_i (1-d)$

$P_o = V_o^2 / R_o$

$\frac{V_i}{(1-d)} = V_o$

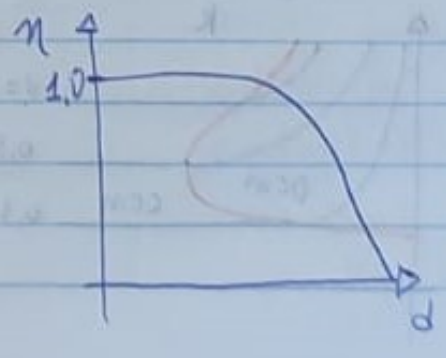
$$\eta = \frac{\frac{V_o^2}{R_o}}{\frac{V_o^2}{R_o} + \left[\frac{V_i}{R_o (1-d)^2}\right]^2 \cdot RL}$$

$$\eta = \frac{\frac{\left(\frac{V_i}{(1-d)}\right)^2}{R_o}}{\frac{\left(\frac{V_i}{(1-d)}\right)^2}{R_o} + \left(\frac{V_i}{R_o (1-d)^2}\right)^2 \cdot RL}$$

→ $\eta = \frac{1}{1 + \frac{RL}{R_o} \frac{1}{(1-d)^2}}$

$$\eta = \frac{1}{1 + \frac{\alpha}{(1-d)^2}}$$

$G = \frac{1}{(1-d)} \cdot \eta$; $\alpha = RL/R_o$



→ INDUTÂNCIA CRÍTICA

$\Delta I_L = I_M - I_m$, EM CONDUÇÃO CRÍTICA $I_m = 0$

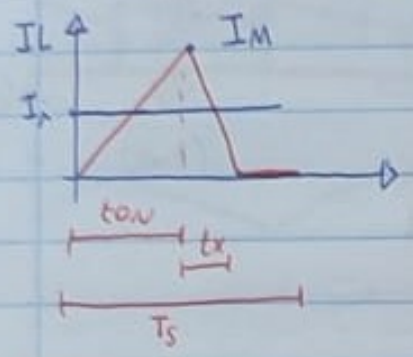
$$\cancel{\Delta I_L = I_M \rightarrow I_M \Delta I_L = I_L + \frac{\Delta I_L}{2} \rightarrow I_M = \frac{I_o}{(1-d)} + \frac{\Delta I_L}{2}}$$

$$I_o = \frac{I_M + I_m}{2}(1-d) \rightarrow I_M = \frac{2I_o}{(1-d)} \quad ; \quad \Delta I_L = \frac{V_i d}{f_s L_{CR}}$$

$$\Delta I_L = I_M$$

$$\frac{2I_o}{(1-d)} = \frac{V_i d}{f_s L_{CR}} \rightarrow L_{CR} = \frac{V_i d(1-d)}{2 I_o f_s}$$

→ MODO DE CONDUÇÃO DESCONTÍNUA



$$I_{L(AVG)} = I_i = \frac{1}{T_s}\frac{I_M(t_{on}+t_x)}{2}$$

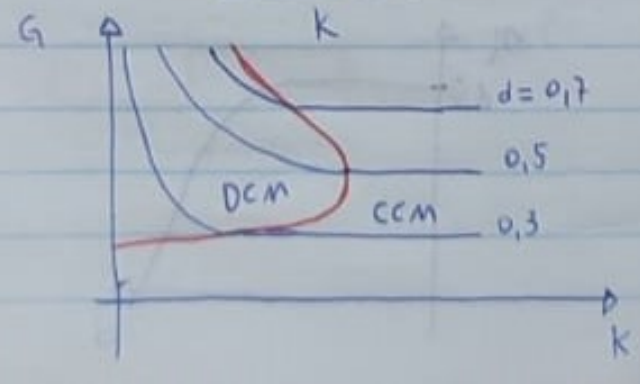
$$t_x = T_s(2I_i - I_M d)/I_M$$

$$t_x = \frac{2I_o T_s}{I_M} \quad ; \quad I_M = \frac{V_i d}{f_s L} \quad ; \quad t_x = \frac{2I_o L}{V_i d}$$

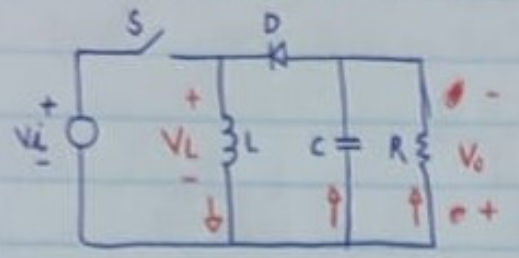
$$\frac{V_o}{V_i} = \frac{dT_s + t_x}{t_x} \rightarrow \frac{V_o}{V_i} = 1 + \frac{V_i d^2}{2 I_o L f_s} \quad ; \quad K = \frac{2I_o L}{V_i T_s}$$

$$G(DCM) = \frac{V_o}{V_i} = 1 + \frac{d^2}{K} \quad ; \quad G[CCM] = \frac{1}{1-d} \rightarrow d = 1 - \frac{1}{M}$$

$$G_{BCM} = \frac{1 + \left(1 - \frac{1}{G_{BCM}}\right)^2}{K} \rightarrow K_F = \frac{G_{BCM} - 1}{G_{BCM}^2}$$

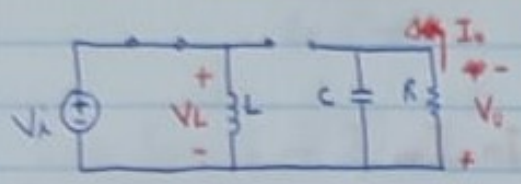


→ CONVERSOR BUCK-BOOST



* OPERAÇÃO EM CCM
* DOIS ESTADOS TOPOLÓGICOS
* $v_L\big|_{t=T_s} = 0$; * $\langle i_C \rangle = 0$
* ELEVADA CAPACITÂNCIA * ELEVADA INDUTÂNCIA

→ PRIMEIRA ETAPA DE OPERAÇÃO ($t_{ON} = dT_s$) → S=ON, D=OFF



$v_L = V_i \rightarrow \dfrac{di_L}{dt} = \dfrac{V_i}{L}$

$i_C = -I_o$

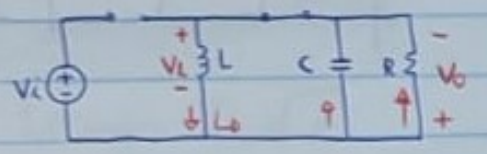
$\hookrightarrow \dfrac{dv_C}{dt} = \dfrac{-v_C}{RC}$

$v_C = V_o$

$$\begin{bmatrix} di_L/dt \\ dv_C/dt \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & -1/RC \end{bmatrix} \begin{bmatrix} i_L \\ v_C \end{bmatrix} + \begin{bmatrix} 1/L \\ 0 \end{bmatrix} \begin{bmatrix} V_i \end{bmatrix} \qquad Y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} i_L \\ v_C \end{bmatrix}$$

$$\dot{x} = A \quad x + B \quad U$$

→ SEGUNDA ETAPA DE OPERAÇÃO ($t_{OFF} = (1-d)T_s$) → S=OFF, D=ON



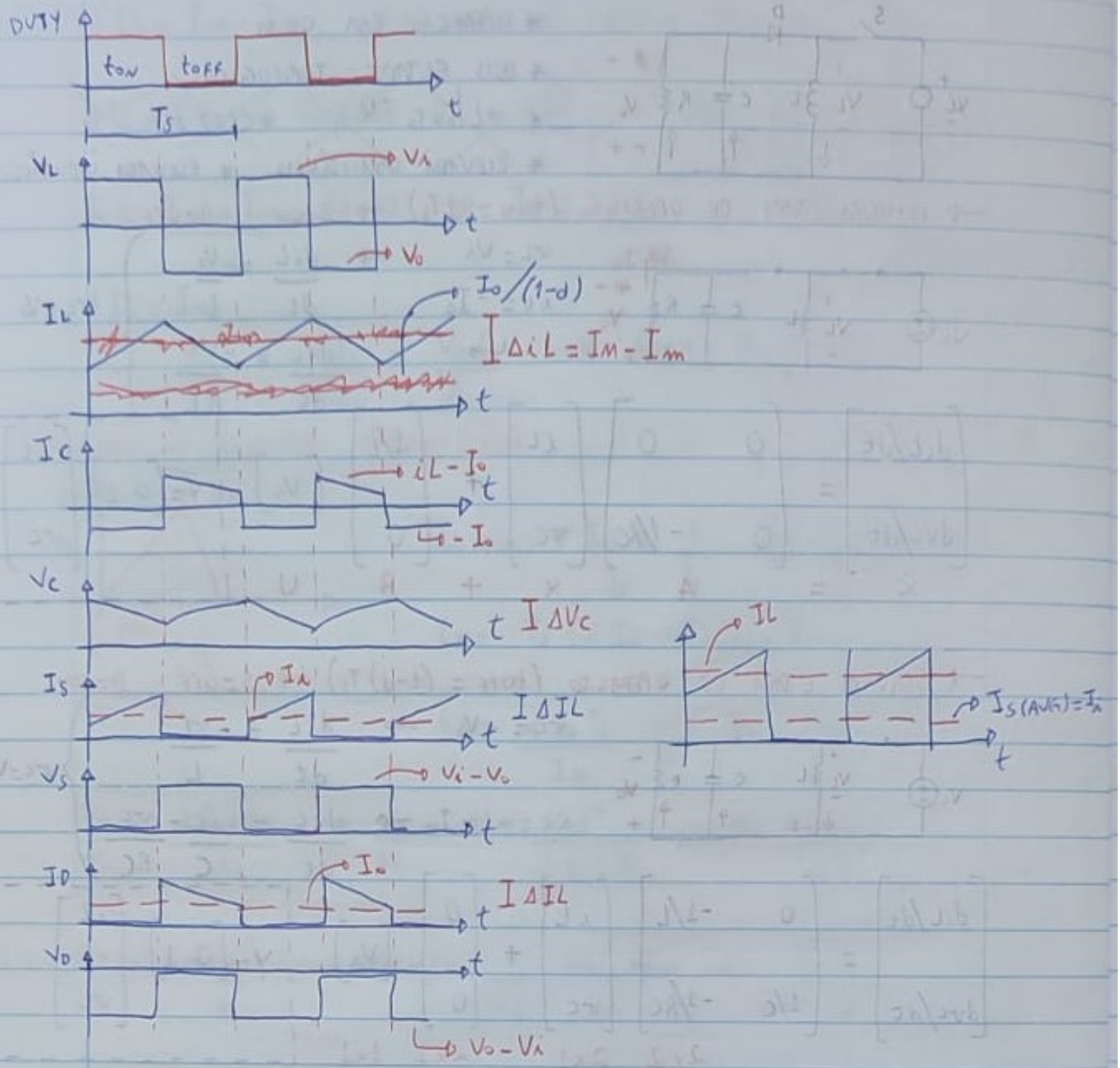
$v_L = -V_o \rightarrow \dfrac{di_L}{dt} = \dfrac{-v_C}{L}$

$i_C = i_L - I_o \rightarrow \dfrac{dv_C}{dt} = \dfrac{i_L}{C} - \dfrac{v_C}{RC}$

$v_C = V_o$

$$\begin{bmatrix} di_L/dt \\ dv_C/dt \end{bmatrix} = \underset{2\times 2}{\begin{bmatrix} 0 & -1/L \\ 1/C & -1/RC \end{bmatrix}} \underset{2\times 1}{\begin{bmatrix} i_L \\ v_C \end{bmatrix}} + \underset{2\times 1}{\begin{bmatrix} 0 \\ 0 \end{bmatrix}} \underset{1\times 1}{\begin{bmatrix} V_i \end{bmatrix}} \qquad Y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} i_L \\ v_C \end{bmatrix}$$

→ FORMAS DE ONDA



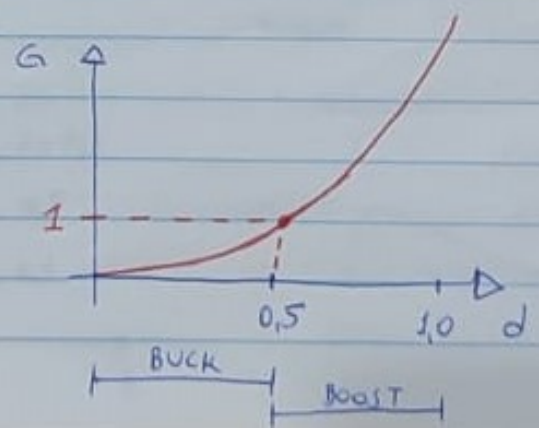
→ GANHO ESTÁTICO

$v_L(ON) = v_L(OFF)$

$V_i \, d \, T_s = V_o (1-d) T_s$

$V_i d - V_o (1-d) = 0$

$$\frac{V_o}{V_i} = \frac{d}{(1-d)}$$



→ ESFORÇOS NOS COMPONENTES

* $P_i = P_o \rightarrow V_i I_i = V_o I_o \rightarrow I_i = \frac{V_o}{V_i} I_o \rightarrow I_i = \frac{I_o d}{(1-d)}$

$I_i = I_o \frac{d}{(1-d)}$; $I_o = I_i \frac{(1-d)}{d}$

↳ $I_i = \frac{V_o d}{R_o (1-d)}$; $V_o = \frac{V_i d}{(1-d)}$; $I_i = \frac{V_i d^2}{R_o (1-d)^2}$

* $I_L = I_S + I_D = I_i + I_o = I_o \frac{d}{(1-d)} + I_o = \underbrace{\frac{I_o}{(1-d)}} = \frac{I_i}{d}$

$I_L = \frac{I_i}{d} = \frac{V_i d}{R_o (1-d)^2}$

* $I_S = I_i = \frac{V_i d^2}{R_o (1-d)^2}$ * $I_{D(AVG)} = I_o = I_i \frac{(1-d)}{d} = \frac{V_o}{R_o}$

$= I_o \frac{d}{(1-d)} = I_L (d) = I_L (1-d)$

* $I_{S(RMS)} = I_L \sqrt{d}$ * $I_{D(RMS)} = I_L \sqrt{(1-d)}$

* $i_{Co(RMS)} = I_o \sqrt{\frac{d}{1-d}}$ * $i_{Co(RMS)} = \sqrt{I_{D(RMS)}^2 - I_{D(AVG)}^2}$

* $V_{S(MAX)} = V_i - V_o$ * $V_{D(MAX)} = V_o - V_i$

→ ESPECIFICAÇÃO DOS ELEMENTOS PASSIVOS

* INDUTOR P/ $t_{ON} = dT_S$

$v_L = L \frac{di}{dt}$

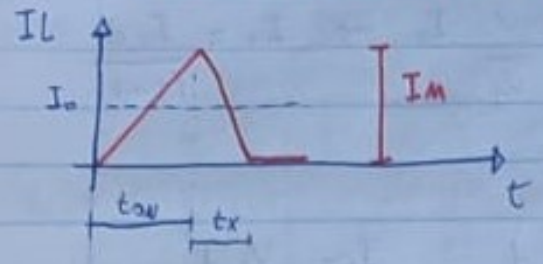
$\boxed{L = \frac{V_i d}{f_S . \Delta I_L}}$

* CAPACITOR EM $t = dT_S$

$i_C = C \frac{dv_C}{dt}$

$\boxed{C = \frac{I_o d}{f_S . \Delta V_C}}$

→ MODO DE CONDUÇÃO DESCONTÍNUA (DCM)



$$V_i\, d\, T_s - V_o\, t_x = 0$$

$$\frac{V_o}{V_i} = \frac{t_{on}}{t_x} \quad (1)$$

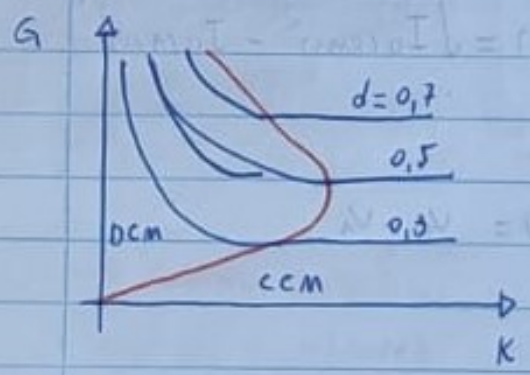
$$I_{L(AVG)} = I_o = \frac{1}{T_s}\,\frac{I_M}{2}\, t_x \rightarrow t_x = \frac{2 I_o}{f_s I_M} \quad (2)$$

$$I_M = \frac{V_i . d}{f_s L} \quad (3)$$

LEVANDO (3) e (2) EM (1) CHEGA-SE EM,

$$\frac{V_o}{V_i} = \frac{V_i\, d^2}{2 I_o L_o f_s} = \frac{d^2}{K} \quad (4)$$

$$K = \frac{2 I_o L_o}{V_i T_s} \quad (5)$$



$$G_{(CCM)} = \frac{d}{1-d} \rightarrow d = \frac{G}{1+G} \quad (6)$$

SUBSTITUINDO (6) EM (4)

$$K_F = \frac{G^2}{G\,(1+G)^2} \quad (7)$$

↳ REGIÃO DE FRONTEIRA ENTRE OS DOIS MODOS DE CONDUÇÃO CCM E DCM.